NOTE: M appears because in FEM you are not working in ordinary $\mathbb{R}^n$ with the standard dot product; you are approximating the Hilbert space $L^2(M, d\mathrm{vol}_g)$. The **mass matrix** M is exactly the finite-dimensional version of that inner product.

Continuous picture

In your notes, states live in

$$\mathcal{H} = L^2(M, d\mathrm{vol}_g),$$

with inner product

$$\langle u, v \rangle = \int_M \bar{u}(x)\ v(x)\, d\mathrm{vol}_g(x).$$

When you expand a wave function in FEM basis functions $\{\phi_i\}$,

$$u_h(x) = \sum_i u_i\, \phi_i(x),$$

the inner product becomes

$$\langle u_h, v_h \rangle = \sum_{i,j} \bar{u_i}\, v_j \int_M \phi_i\, \phi_j\, d\mathrm{vol}_g.$$

The integrals $\int_M \phi_i\, \phi_j\, d\mathrm{vol}_g$ are exactly the entries of the mass matrix:

$$M_{ij} = \int_M \phi_i\,(x) \phi_j(x)\, d\mathrm{vol}_g(x).$$

So M is the matrix that tells you how to compute norms and inner products of discrete states in a way that still respects the manifold's metric.

Discrete eigenproblem

The continuous eigenproblem is

$$\hat{H}\psi = E\psi.$$

Projecting onto the FEM basis and using the L^2 inner product gives

$$\sum_j H_{ij}\, u_j = E \sum_j M_{ij}\, u_j,$$

so in matrix form:

$$Hu = EMu.$$

That's why a **generalized** eigenvalue problem appears. $H$is the matrix representation of your Hamiltonian in the $\{\phi_i\}$basis, and $M$is the representation of the inner product.file

If you instead forced $M = I$(standard dot product on coefficients), you would be implicitly changing the Hilbert space and losing the correct geometry and volume form $d\mathrm{vol}_g$that your notes spend time defining.file

Local FEM view

On each triangle T, for P1 elements,

$$M_T = \int_T \begin{pmatrix} \phi_1^2 & \phi_1\phi_2 & \phi_1\phi_3 \\ \phi_2\phi_1 & \phi_2^2 & \phi_2\phi_3 \\ \phi_3\phi_1 & \phi_3\phi_2 & \phi_3^2 \end{pmatrix} dS = \frac{|\,T\,|}{12}\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Assembling these local blocks over all triangles gives the global mass matrix M.file

So Mappears because:

- you expanded ψ in non-orthonormal FEM basis functions, and
- you want your discrete problem to live in the same geometric L^2 space your continuous theory is written in.

That's why the proper discrete Hamiltonian eigenproblem is $Hu = EMu$, not just $Hu = Eu$.

For our chosen potential we'll be taking into consideration the geodesic harmonic trap

A geodesic harmonic trap is just the curved-space version of the ordinary harmonic oscillator: instead of using Euclidean distance from a center, you use **geodesic distance along the surface**. So, the potential is

$$V(x) = \frac{k}{2}\, d_g(x, x_0)^2,$$

where x_0 is the trap center and d_g is the intrinsic distance on the manifold. This is the natural oscillator-style potential on a curved space and is commonly written in geodesic coordinates for constant-curvature spaces.

What each symbol means

- x: the point where you evaluate the potential.
- x_0: the point on the surface where the trap is centered.
- $d_g(x, x_0)$: shortest distance **along the surface**, not through ambient $\mathbb{R}^3$.
- k: trap strength; bigger k means tighter confinement.

So near x_0, the particle sees a bowl-shaped potential, just like the flat harmonic oscillator, except the distance is measured intrinsically on the manifold.

Sphere version

Since we're working on a sphere, the geodesic distance between two points x and x_0 on a sphere of radius R is the great-circle distance

$$d_g(x, x_0) = R \arccos\left(\frac{x \cdot x_0}{R^2}\right).$$

That is the standard spherical-distance formula.

So, our potential becomes

$$V(x) = \frac{k}{2} R^2 \arccos^2\left(\frac{x \cdot x_0}{R^2}\right).$$

The source term f: definition and structure

We'll take f to be:

$$f(x) = A\, e^{i\ell\varphi(x)} \exp\left(-\frac{d_g(x, x_0)^2}{\sigma^2}\right)$$

$x \in S^2 \subset \mathbb{R}^3$.

$x_0 \in S^2$ is the center (you'll probably take x_0 as the north pole).

$d_g(x, x_0)$is the geodesic distance on the sphere:

$$d_g(x, x_0) = \arccos(x \cdot x_0) \text{ on the unit sphere.}$$

$\varphi(x)$is the azimuthal angle (longitude). In spherical coordinates (θ, φ), this is your usual φ(adjust for your swapped θ, ϕconvention).

$A \in \mathbb{C}$ (or $\mathbb{R}$) is an amplitude.

$\ell \in \mathbb{Z}$ controls the angular momentum content.

$\sigma > 0$ sets the geodesic width of the Gaussian.

Decomposition into triangle contributions

Let the mesh be a triangulation $\mathcal{T}_h$of (a discrete approximation of) S^2. Then

$$F_i = \sum_{T \in \mathcal{T}_h} \int_T f(x)\ \phi_i(x)\, d\mathrm{vol}_g(x),$$

and for each triangle $T$with vertices (x_a, x_b, x_c)you have local basis functions ϕ_a, ϕ_b, ϕ_c(the nodal hats restricted to T).doc.freefem+1

For a fixed triangle T, write the local load vector

$$F^T = \begin{pmatrix} F_a^T \\ F_b^T \\ F_c^T \end{pmatrix}, F_a^T = \int_T f\ \phi_a\, d\mathrm{vol}_g, \text{ etc.}$$

Then

$$F_i = \sum_{T \ni i} F_i^T.$$

So the entire discrete RHS is built by looping over triangles and adding these local contributions to the global indices, exactly as you do for stiffness/mass.

On your sphere, you approximate $d\mathrm{vol}_g$by the Euclidean area of the embedded triangle so numerically your M_T has exactly this form with area as parameter.

Now,

$$F_p^T = \int_T f(x)\ \phi_p(x)\, d\mathrm{vol}_g(x).$$

If f were constant on T, say $f(x) \approx f_T$for all $x \in T$, then

$$F_p^T \approx f_T \int_T \phi_p\,(x)\, d\mathrm{vol}_g(x).0$$

But

$$\int_T \phi_p\,(x)\, d\mathrm{vol}_g(x) = \sum_{q\in\{a,b,c\}} \int_T \phi_p\,(x)\phi_q(x)\, d\mathrm{vol}_g(x) \cdot 1 = \sum_q (\,M_T)_{pq},$$

since $\phi_a + \phi_b + \phi_c \equiv 1$on T.

Therefore

$$F_p^T \approx f_T \sum_q (\,M_T)_{pq},$$

or, vectorially,

$$F^T \approx M_T\mathbf{1}\quad f_T,$$

where $\mathbf{1} = (1,1,1)^\top$.

This is exactly the pattern: the local load vector is the local mass matrix times a vector of ones, scaled by an appropriate sample of $f$on the triangle.

FINDING THE ANALYTIC SOLUTION USING GEOMETRIC FOURIER TRANSFORM

If you were to guess the Fourier transform derived from the geometry of the sphere, you'd say something like:

$$\mathcal{F}[f](\omega) = \int_{S^2} f(x)e^{-2\pi i\langle x,\omega\rangle_g} dvol_g$$

The problem with this formula is that it's not exactly why the Fourier transform works on PDEs in the first place.

The notion of defining the usual Fourier transform was that $\hat{f}(\omega)$ was the projection of f onto the eigen basis of $\frac{d}{dx}$, as a linear operator. The thing is that the exponential here is not an eigenfunction of our differential Δ_g

In standard spherical coordinates (θ, φ), with ϕ= colatitude, θ= longitude,

$$\Delta_g u = \frac{1}{\sin^2 \phi} \frac{\partial^2 u}{\partial \theta^2} + \frac{1}{\sin \phi} \frac{\partial}{\partial \phi} (\sin \phi \frac{\partial u}{\partial \phi})$$

look for eigenfunctions

$$-\Delta_g Y(\theta, \varphi) = \lambda Y(\theta, \varphi).$$

Try separated variables: $Y(\theta, \varphi) = \Theta(\theta)\Phi(\varphi)$. Plugging in and separating gives two ODEs:

Azimuthal Equiation:

$$\frac{1}{\Phi} \frac{d^2\Phi}{d\varphi^2} = -m^2 \Rightarrow \Phi(\varphi) = e^{im\varphi}$$

Polar equation (In $x = cos(\theta)$):

$$(1 - x^2)\frac{d^2\Theta}{dx^2} - 2x\frac{d\Theta}{dx} + (\ell(\ell + 1) - \frac{m^2}{1 - x^2})\Theta = 0$$

whose solutions are the **associated Legendre functions** $P_\ell^m(x)$

Putting the pieces together,

$$Y_\ell^m(\theta, \varphi) = N_{\ell m} P_\ell^m(\cos \theta) e^{im\varphi},$$

with a normalization constant chosen so that

$$\int_{S^2} \bar{Y}_\ell^m Y_{\ell'}^{m'} d\Omega = \delta_{\ell\ell'}\delta_{mm'}.$$

A standard choice is

$$Y_\ell^m(\theta, \varphi) = \sqrt{\frac{(2\ell + 1)(\ell - m)!}{4\pi(\ell + m)!}} P_\ell^m(\cos \theta) e^{im\varphi}.$$

So now we project onto the eigen-basis of our operator to get

$$\mathcal{F}[f](\ell, m) = \int_{S^2} f(\theta, \varphi)\overline{Y_\ell^m(\theta, \varphi)}\ dvol_g,$$

Is F well-defined?

In other words, if we define the map

$$\mathcal{F}: L^2(S^2) \to \ell^2(\{(\ell, m)\})$$

$$f \to (\hat{f}_{lm}) = \left(\int_{S2} f(\theta, \varphi)Y_\ell^m(\theta, \varphi)dvolg\right)$$

Do we get a well-defined map?

And one can see using basic integration theory and Plancherel that we do have a well-defined map as well as

$$\| f \|_{L^2}^2 = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \left|\hat{f}_{\ell m}\right|^2$$

For $s \geq 0$, the Sobolev space $H^s(S^2)$ can be described via the spherical harmonic expansion. A standard equivalent definition is:

- $f \in H^s(S^2)$iff $f \in L^2(S^2)$and

$$\| f \|_{H^s(S^2)}^2 \sim \sum_{\ell=0}^{\infty} (1 + \ell(\ell + 1))^s\, S_{ff}(\ell),$$

where $S_{ff}(\ell) = \sum_{m=-\ell}^{\ell} | \hat{f}_{\ell m} |^2$is the power spectrum at degree ℓ.

In other words, the Sobolev norm of order $s$is equivalent to a weighted ℓ^2-norm of the spherical harmonic coefficients, with weights $(1 + \lambda_\ell)^s$where $\lambda_\ell = \ell(\ell + 1)$are the Laplace–Beltrami eigenvalues.

This shows explicitly:

$\mathcal{F}$ is a linear isomorphism between $H^s(S^2)$and the sequence space

$$\left\{\hat{f}_{\ell m}: \sum_{\ell,m} (1+\ell(\ell+1))^s \mid \hat{f}_{\ell m} \mid^2 < \infty\right\}.$$

The norm on $H^s(S^2)$is equivalent to the norm on this coefficient space.

So let's apply the transform to our strong solution

The strong-form equation is

$$-\frac{\hbar^2}{2m}\Delta_g u + Vu = f.$$

Apply the spherical harmonic transform: take inner products with $\overline{Y_\ell^m}$on both sides. Using orthonormality, you get for each (ℓ, m):

$$\frac{\hbar^2}{2m}\lambda_\ell\, u_{\ell m} \;+\; \int_{S^2} Vu\, \overline{Y_\ell^m}\, \mathrm{dvol}_g \;=\; f_{\ell m}.$$

If $V$is **constant** (say $V \equiv V_0$), then

$$\int_{S^2} V_0\, u\, \overline{Y_\ell^m} dvol_g = V_0 u_{\ell m},$$

and the mode equation simplifies to

$$\left(\frac{\hbar^2}{2m}\lambda_\ell + V_0\right) u_{\ell m} = f_{\ell m}.$$

Solving one would get:

$$u_{\ell m} = \frac{f_{\ell m}}{\frac{\hbar^2}{2m}\ell(\ell+1) + V_0}.$$

So our solution would yield

$$u(\theta,\varphi) = \sum_{\ell,m} u_{\ell m}\, Y_m^\ell(\theta,\varphi)$$

to test the FEM method more concretely we'll substitute our V, with a particular V_0...

There are two natural ways to compare:

A. Compare in physical space (nodal values)

1. Choose a test mesh refinement level and fix all parameters (A_0, ℓ, σ, V_0).perplexity

2. For each vertex $x_i$of your mesh, evaluate the analytic solution

$$u_{\text{analytic}}(x_i) = \sum_{\ell=0}^{L_{\max}} \sum_{m=-\ell}^{\ell} u_{\ell m}\, Y_\ell^m(\theta, \varphi)$$

truncated at some $L_{\max}$based on your resolution.

3. Compare the FEM nodal values $u_h(x_i) = u_i$to $u_{\text{analytic}}(x_i)$, for example via

$$\max_i \mid u_i - u_{\text{analytic}}(x_i) \mid = \left(\sum_i \mid u_i - u_{\text{analytic}}(x_i) \mid^2 \right)^{1/2}.$$

4. Refine the mesh (increase subdivisions) and check that this error decays roughly like some power of the mesh size h(first order for P1 elements).people.tamu+2

This is the most direct “are my numbers right?” comparison.

B. Compare in spectral space (project FEM onto spherical harmonics)

1. Interpret $u_h$as a function on the sphere (piecewise linear).
2. Approximate its spherical harmonic coefficients numerically:

$$\hat{u}_{\ell m}^{(h)} \approx \int_{S^2} u_h\,(\theta, \varphi)\, Y_\ell^m(\bar{\theta}, \varphi)\, d\text{vol}_g,$$

using a quadrature on the sphere (or sampling at mesh vertices with appropriate weights).

3. Compare $\hat{u}_{\ell m}^{(h)}$to the analytic $u_{\ell m}$:

$$\mid \hat{u}_{\ell m}^{(h)} - u_{\ell m} \mid$$

as ℓ varies.

4. Again, refine the mesh and see the spectral error shrink.

This approach is conceptually neat because it uses the same basis (spherical harmonics) in theory and in your comparison.

To begin implementing the error analysis code (using method B) we'll start by discussing

$$\boxed{f_{\ell m} = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}} \int_0^{2\pi} \int_0^{\pi} f(\theta,\varphi)\, P_\ell^m(\cos\,\theta)\, e^{-im\varphi} \sin\,\theta\, d\theta\, d\varphi.}$$

But For our specific source

$$f(\theta,\varphi) = A\, e^{i\ell_0\varphi}\, \exp\left(-\frac{d_g((\theta,\varphi),x_0)^2}{\sigma^2}\right),$$

the spherical harmonic coefficient integral becomes explicit but not closed-form elementary. Let's write it carefully and see what structure we get.

Then

$$f_{\ell m} = A\, N_{\ell m} \int_0^{2\pi} \int_0^{\pi} e^{i\ell_0\varphi}\, e^{-\theta^2/\sigma^2}\, P_\ell^m(\cos\,\theta)\, e^{-im\varphi} \sin\,\theta\, d\theta\, d\varphi,$$

where $N_{\ell m} = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}}$

Factor the φ-dependence:

$$f_{\ell m} = A\, N_{\ell m} \left(\int_0^{2\pi} e^{i(\ell_0-m)\varphi}\, d\varphi\right)\left(\int_0^{\pi} e^{-\theta^2/\sigma^2}\, P_\ell^m(\cos\,\theta) \sin\,\theta\, d\theta\right).$$

The $\boldsymbol{\varphi}$-integral: selection of $\boldsymbol{m = \ell_0}$

Compute

$$\int_0^{2\pi} e^{i(\ell_0-m)\varphi}\, d\varphi = \begin{cases} 2\pi, & m = \ell_0, \\ 0, & m \neq \ell_0. \end{cases}$$

So you get **exactly one azimuthal index**:

$$f_{\ell m} = 0 \text{ if } m \neq \ell_0,$$

and for $m = \ell_0$,

$$f_{\ell,\ell_0} = 2\pi A\, N_{\ell,\ell_0} \int_0^{\pi} e^{-\theta^2/\sigma^2}\; P_\ell^{\ell_0}(\cos\ \theta)\sin\ \theta\, d\theta.$$

So it only matters for us to solve $u_{\ell m}$ when $m = \ell_0$

$$u_{\ell\ell_0} = \frac{f_{\ell\ell_0}}{\frac{\hbar^2}{2m}\ell(\ell+1) + V_0}$$

And therefore

$$u = \sum_{\ell=0}^{\infty} u_{\ell\ell_0}\, Y_\ell^{\ell_0}(\theta,\varphi)$$

To compare the error of our calculated FEM we decided to switch to a constant potential and then find the L^2 difference between u_h and u. one must be vigilant with the use of the norm, since we're using the mesh's geometry, we compute our global mass matrix and use it as our metric.

So first

$$err_h = u_h - u$$

And then

$$\|err_h\|_{L^2} = \left\|err_h^{\dagger} \cdot (Merr_h)\right\|_2$$